

HW04 - AI Automation Testing Main Report

1. General Information

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- **Assignment:** HW04 - AI Automation Testing
- **Public repository:** [namdin05/ST-Group - branch `23127430-HW04`](#)
- **Self-assessed grade:** 90/100

2. AI Use Declaration

I use AI tools for the following tasks:

- Review human-approved feature test cases and external test data.
- Generate and self-review data-driven Playwright TypeScript automation.
- Inspect rendered selectors, execute multi-browser tests, analyze evidence, and generate reports.
- Preserve genuine AI interactions in the audit report.

3. Selected Features

Pool	Feature ID	Feature name
A	FR-01	Account Registration
B	FR-07	Shopping Cart
C	FR-13	Dashboard

4. Test Environment and Tools

Item	Value
Automation framework	Playwright + TypeScript
Reporting tool	Playwright HTML Reporter
Browsers	Chromium, Firefox, WebKit
SUT environment	Web http://localhost:5173 ; Web Admin http://localhost:5174 ; API http://localhost:3000

Item	Value
Execution environment	Windows / PowerShell; latest execution date 2026-08-11 (Asia/Saigon)

5. Automation Approach

Each feature reads reviewed input and expected results from a dedicated JSON file. UI tests prefer semantic Playwright locators and use state-based synchronization. FR-07 UI tests control only the product API response with reviewed fixtures so that price/quantity preconditions are deterministic; cart behavior itself runs in the real React application. FR-13 UI tests control Admin API responses with reviewed order fixtures while exercising the real React Dashboard. FR-13 API cases run a temporary copy of the reviewed backend source with a private SQLite database and generated users. These strategies avoid ordering dependencies and leave the source SUT unchanged.

6. FR-01 Automation Report

6.1 Feature Overview

- **Feature ID:** FR-01
- **Feature name:** Account Registration

6.2 Test Cases Selected for Automation

All 18 reviewed cases in `automation/test-cases/fr01.md` were automated: UI cases TC001-TC008, API cases TC009-TC017, and UI + API security case TC018.

Coverage is not complete against the feature specification, however: FR-01 requires the password and confirmation password to match, but the selected case set contains no mismatch case.

6.3 Test Data File

`automation/test-data/fr01-registration.json`

6.4 Automation Script

`automation/tests/fr01-registration.spec.ts`

6.5 Assertions Used

Assertions cover required-field behavior, native email semantics, password validation messages, registration/login URL transitions, API status classes and response keys, inability to log in after rejected registration, preservation of the original duplicate account, forced-role handling, and inert rendering of script-like input. Playwright auto-waiting and state-based assertions are used; the script contains no fixed sleeps.

6.6 Browser Execution Results

Browser	PASS	FAIL	Total	HTML report
Chromium	7	11	18	html-reports/fr01/chromium/index.html
Firefox	7	11	18	html-reports/fr01/firefox/index.html
WebKit	7	11	18	html-reports/fr01/webkit/index.html
Total	21	33	54	

PASS on all browsers: TC002, TC004, TC006, TC008, TC009, TC015, TC017.

FAIL on all browsers: TC001, TC003, TC005, TC007, TC010-TC014, TC016, TC018.

The three feature-browser reports above are the canonical FR-01 results. The consolidated diagnostic report at <html-reports/fr01/review-20260809-v2/index.html> shows 20 PASS / 34 FAIL because a 15-second diagnostic timeout and three parallel workers caused an additional Firefox timeout in TC002. The focused rerun at <html-reports/fr01/review-20260809-tc002-firefox/index.html> passed, so the diagnostic timeout is excluded from the primary totals.

6.7 AI-generated Problems Found

Review found the following automation and data-contract problems:

1. TC018 data permits either rejection or escaped-text handling, but the script first requires registration status 200/201. A conforming 4xx rejection would therefore be reported as a false failure.
2. API test data declares `request.method`, but the helper always sends POST; this is currently harmless because every FR-01 API case uses POST, but the data-driven contract is misleading.
3. TC011 and TC012 assert only the 4xx status and do not independently prove the expected "no account created" postcondition. Other negative cases use a login check, but that helper hard-codes 401 instead of accepting the contractually relevant 4xx outcome.
4. TC007 cannot reach its duplicate-email assertion because the UI incorrectly rejects the otherwise valid password first. Its current failure is inconclusive for the intended duplicate-email behavior.
5. TC016 proves that the duplicate request returns the wrong status and that the original account remains usable, but it does not independently query storage to prove that no second record was created.

6.8 Human Revisions

The source cases are marked Reviewed/Revised, but the abnormalities identified in this review have not yet been human-approved or corrected. No new human revision is claimed.

6.9 Genuine SUT Bugs, if any

Nine FR-01 defects have been human-reviewed and documented in `bug_report.md` with public GitHub Issues: [#18](#), [#19](#), [#38](#), [#39](#), [#40](#), [#41](#), [#42](#), [#43](#), and [#46](#).

The canonical automation also provides the following reproducible evidence across all three browsers:

- TC001 and TC005: the confirmation-password field is absent, and valid strong passwords containing an allowed special character are rejected by the UI.
- TC003: the email control uses `type="text"`, and no invalid-email message is shown.

- TC010: the API accepts a missing name and creates a login-capable account.
- TC011 and TC012: the API returns 200 for missing email or missing password instead of a 4xx client error.
- TC013: the API accepts an invalid email and creates a login-capable account.
- TC014: the API accepts a seven-character password and creates a login-capable account.
- TC016: the API returns 200 for a duplicate email.
- TC018: a script-like name is accepted and rendered in the header as a `<script>` element rather than inert text.

TC007 is not counted as separate duplicate-email automation evidence because the upstream password-validation failure prevents that UI behavior from being exercised. Any failing behavior not represented by one of the nine reviewed issues remains a candidate.

6.10 Limitations, if any

- The FR-01 specification explicitly requires confirmation-password matching, while the note in `fr01.md` says confirmation is not required and TC006 replaced the mismatch case. This is a requirement/test-case contradiction and a real coverage gap.
- Status and Actual Result fields in `fr01.md` are stale relative to the canonical reports: several cases remain PAUSED, while TC018 is marked PASSED although it fails on all three browsers.
- The rendered registration inputs have no accessible name, id, or placeholder, so the page object uses form-scoped positional selectors (`nth(0)/nth(1)`); these are brittle if field order changes.
- Unique run IDs isolate parallel tests and duplicate fixtures, but the SUT has no cleanup API, so generated accounts remain in the database.
- TC008 is exploratory: PASS means the page remained responsive and did not insert a form script; it does not prove an unspecified maximum length or mandatory rejection policy.
- TC017 assumes that ignoring the submitted role and creating a normal user is the only acceptable path. The specification does not clarify whether rejecting the unexpected role with 4xx should also pass.
- SEC-01 plaintext-password storage is not verified by this FR-01 suite.

7. FR-07 Automation Report

7.1 Feature Overview

- **Feature ID:** FR-07
- **Feature name:** Shopping Cart

7.2 Test Cases Selected for Automation

All 14 reviewed cases in `automation/test-cases/fr07.md` were automated: UI cases TC001-TC010 and API cases TC011-TC014.

7.3 Test Data File

automation/test-data/fr07-shopping-cart.json

7.4 Automation Script

automation/tests/fr07-shopping-cart.spec.ts

7.5 Assertions Used

Assertions cover semantic table columns, empty-state content and illustration, row count, quantity boundaries, line/grand-total arithmetic, exact total and navigation labels, confirmation-dialog branches, URL transitions, API status/body shape, duplicate merge behavior, and unchanged cart state after invalid quantities.

7.6 Browser Execution Results

Browser	PASS	FAIL	Total	HTML report
Chromium	3	11	14	html-reports/fr07/chromium/index.html
Firefox	3	11	14	html-reports/fr07/firefox/index.html
WebKit	3	11	14	html-reports/fr07/webkit/index.html
Total	9	33	42	

PASS on all browsers: TC006, TC011, TC013.

FAIL on all browsers: TC001-TC005, TC007-TC010, TC012, TC014.

The full Firefox run had a one-time page crash during TC008 setup. A focused Firefox rerun completed normally and reproduced the expected failed assertions (missing confirmation dialog and immediate removal). The rerun is stored at `html-reports/fr07/firefox-tc008-rerun/index.html`.

7.7 AI-generated Problems Found

Two genuine automation problems were found during the first validation runs:

1. The add-to-cart locator used the fixed accessible name Thêm vào giỏ hàng; it stopped resolving after the button changed to Đã thêm.
2. Setup used `page.goto()` between product/cart routes, which reloaded the React application and reset the in-memory CartProvider state.

Both invalid runs were excluded. The locator now accepts both observable button states, and setup navigates through real SPA links to preserve cart state. TypeScript compilation passed after revision.

7.8 Human Revisions

Pending human review. The two problems above were AI self-review corrections before the final recorded execution; no human revision is claimed.

7.9 Genuine SUT Bugs, if any

Seven FR-07 defects have been human-reviewed and documented in `bug_report.md` with public GitHub Issues: [#20](#), [#21](#), [#22](#), [#44](#), [#45](#), [#50](#), and [#51](#).

The final runs produced the following 11 reproducible failure cases; behaviors without a matching reviewed issue remain candidates:

- TC001: header is Giá, not required Đơn giá.
- TC002: empty cart has no illustration.
- TC003 and TC012: duplicate products create two lines instead of merging quantities.
- TC004 and TC005: cart rows have no +/- quantity controls.
- TC007: label is Tổng tạm tính, not Tổng cộng.
- TC008 and TC009: remove action has no confirmation dialog; cancel branch cannot be performed.
- TC010: non-empty cart uses ← Mua tiếp, not Tiếp tục mua sắm.
- TC014: API returns 200, stores quantity 0/-1, and changes the cart.

7.10 Limitations, if any

- The prompt says FR-01 once, but every supplied artifact and the execution request identify FR-07; automation follows FR-07.
- Reviewed product prices differ from the current SUT database. UI tests mock only `/api/products*` with the reviewed JSON fixtures; this must be approved as the intended deterministic precondition strategy.
- The product quantity input lacks an associated accessible label, so one scoped `input[type="number"]` fallback is documented in the script.
- The SUT provides no test cleanup API. Unique test users prevent cross-test collisions but remain in the SQLite database.
- TC011 verifies authenticated 200 plus array shape for a fresh isolated user; an empty array alone cannot independently prove ownership semantics.

8. FR-13 Automation Report

8.1 Feature Overview

- **Feature ID:** FR-13
- **Feature name:** Dashboard

8.2 Test Cases Selected for Automation

All 14 reviewed cases in `automation/test-cases/fr13.md` were automated: UI cases TC001-TC004 and TC009-TC011, API cases TC005-TC008 and TC012-TC013, and UI + API consistency case TC014.

8.3 Test Data File

automation/test-data/fr13-dashboard.json

8.4 Automation Script

automation/tests/fr13-dashboard.spec.ts

8.5 Assertions Used

Assertions cover delivered-only revenue aggregation, total order count, zero/one/large-number boundaries, currency and locale-independent thousands grouping, login-only unauthenticated state, complete Admin order fixtures, missing-token/user-role authorization, numeric precision, empty API responses, and equality between the API response consumed by the UI and the rendered metrics. Money values use state-based polling so the assertion cannot pass or fail against React's initial 0 ¤ render.

8.6 Browser Execution Results

Browser	PASS	FAIL	Total	HTML report
Chromium	8	6	14	html-reports/fr13/chromium/index.html
Firefox	8	6	14	html-reports/fr13/firefox/index.html
WebKit	8	6	14	html-reports/fr13/webkit/index.html
Total	24	18	42	

PASS on all browsers: TC002, TC003, TC005-TC008, TC011, TC013.

FAIL on all browsers: TC001, TC004, TC009, TC010, TC012, TC014.

Each canonical report contains Run by: 23127430, a real ISO execution timestamp, 8 expected / 6 unexpected results, and retained traces for all six failures. Five UI failures also contain screenshots; the API-only TC012 failure has a trace and error context.

8.7 AI-generated Problems Found

Three automation/data problems were found and corrected before the canonical runs:

1. TC004 test data hard-coded 1.000.000.000 ¤, although the reviewed case requires thousands grouping but does not prescribe a locale. It now records a grouping requirement and accepts standard dot, comma, regular-space, non-breaking-space, or narrow-space separators.
2. The first money helper verified ¤ and then read the number once. Because React initially renders 0 ¤, a conforming non-zero result could race with the asynchronous state update. It now polls the numeric UI state until the expected value or assertion timeout.
3. The first API assertions checked only response length and delivered sum, which could miss wrong non-delivered rows. They now compare the complete reviewed order projection; negative authorization cases explicitly reject an order array.

The pre-synchronization Chromium run is excluded. The final Chromium rerun and the Firefox/WebKit runs use the corrected script.

8.8 Human Revisions

Pending human review. The three problems above were AI self-review corrections before the final recorded execution; no human revision is claimed.

8.9 Genuine SUT Bugs, if any

The doubled-revenue behavior is human-reviewed and documented as [GitHub Issue #23](#). Six cases provide reproducible evidence on all three browsers, representing one confirmed defect and one remaining candidate:

- TC001, TC004, TC009, TC010, and TC014: the Dashboard doubles every delivered `total_amount`. Examples include expected/actual `350,000/700` `đ`, `1/2` `đ`, and `1,000,000,000/2,000,000,000` `đ`. Order counts and exclusion of non-delivered statuses are otherwise consistent with the controlled response.
- TC012: a valid normal-user JWT receives 200 and the Admin orders array instead of 403 with no Admin data. This remains a candidate because no matching human-reviewed issue is recorded in `bug_report.md`.

The confirmed revenue defect and unconfirmed authorization candidate are kept distinct in the totals.

8.10 Limitations, if any

- Dashboard metric values have no test ID or ARIA relationship. The page object anchors on the verified semantic heading and scopes to the sibling value paragraph in the same card.
- UI metric tests mock the Admin API with the human-reviewed fixtures. They exercise the real React rendering and calculation but do not mutate the live database.
- API data-boundary tests run the reviewed backend source from a temporary directory with a private SQLite database. Bootstrap deliberately fails if the source anchors change, so the harness must be reviewed after backend refactoring.
- TC014 compares the UI with the exact controlled response consumed by that page; it is not an end-to-end comparison against the deployed live database.
- `fr13.md` contains stale manual Actual Result/Status values relative to these canonical runs. The source test-case file was not changed in this script/test-data-only step.

9. Data-driven Testing

Each selected feature will read reviewed test data from its own JSON file. Credentials, tokens, personal data, and unreviewed expected results must not be stored in these files.

Status: implemented for FR-01, FR-07, and FR-13.

10. Assertion Patterns

- Exact status and URL assertions.
- Semantic visibility/count/text/attribute assertions.
- Numeric money assertions after removing display separators.

- Soft assertions where multiple independent contract violations should be captured in one report.
- Deep equality for unchanged API cart state.

11. Multi-browser Execution

Feature	Chromium	Firefox	WebKit
FR-01	7 PASS / 11 FAIL	7 PASS / 11 FAIL	7 PASS / 11 FAIL
FR-07	3 PASS / 11 FAIL	3 PASS / 11 FAIL	3 PASS / 11 FAIL
FR-13	8 PASS / 6 FAIL	8 PASS / 6 FAIL	8 PASS / 6 FAIL

12. Human Review and AI Gap Analysis

ID	Feature	Initial AI output	Problem found	Why it was a problem	Human revision	Retest result
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GAP-01	FR-07	Fixed-name add button locator	Locator detached after text changed to Thêm	UI setup failed before business assertions	None; AI self-corrected to a two-state semantic locator	Setup passed in final Chromium/Firefox/WebKit runs
GAP-02	FR-07	page.goto() used for all setup navigation	Full reload reset in-memory cart state	Produced false failures with an empty cart	None; AI self-corrected to SPA link navigation	Cart preconditions persisted in final runs
GAP-03	FR-01	Case note says confirmation password is out of scope	Contradicts the explicit FR-01 requirement and removes the mismatch scenario	Required behavior is not covered by any selected case	Pending human review	Not retested
GAP-04	FR-01	TC018 allows rejection or escaped text in JSON	Script requires 200/201 before checking the allowed handling	A correct 4xx rejection would be a false failure	Pending human review	Not retested
GAP-05	FR-01	TC007 expects a duplicate-email message after UI submission	Valid test password is rejected before the duplicate request is made	Failure cannot establish duplicate-email behavior	Pending human review	Not retested
GAP-06	FR-13	TC004 data fixed the separator to .	toLocaleString() is browser-locale dependent, while the reviewed case requires grouping rather than one separator glyph	A valid comma/space-formatted value would be a false failure	None; AI replaced the literal with a locale-independent grouping contract	Corrected assertion executed in all three canonical runs
GAP-07	FR-13	Money helper read the number once after finding đ	Initial đ render could race with the Admin orders state update	Could produce a false result before the UI settled	None; AI changed the numeric assertion to state-based polling	Stable expected/actual values reproduced on all three browsers
GAP-08	FR-13	API cases checked only row count and delivered sum	Wrong non-delivered rows could still pass	Source-data assertions were weaker than the reviewed fixtures	None; AI compares the complete order projection	All positive API fixture cases passed on all three runs

13. Test Cases That Could Not Be Automated

All selected cases were automated: 18 for FR-01, 14 for FR-07, and 14 for FR-13. FR-01 confirmation-password mismatch behavior could not be automated from the selected set because no such test case exists; it requires a reviewed case/data addition.

14. Genuine Bug Reports

The selected features have **17 human-reviewed GitHub Issues** documented in ``bug_report.md``: FR-01 has 9, FR-07 has 7, and FR-13 has 1. The report includes reproduction steps, expected/actual results, severity/priority, environment, and direct issue links. Remaining failing behaviors are not counted as genuine bugs until a matching human-reviewed issue exists.

15. Demo Video

- **Task 2 - end-to-end automation demonstration:** <https://youtu.be/8xCs55BQZdl>
- The link was supplied by the student. Duration, narration, and the required authorship evidence should be checked during submission review.

16. Agent Skill

- **Automation skill:** `agent-skill/SKILL.md`
- **AI audit skill:** `audit-skill/SKILL.md`
- **Demonstration:** FR-07 and FR-13 interactions record script generation, selector validation, AI self-review corrections, multi-browser execution, report verification, and audit logging.
- **Agent Skill video:** <https://youtu.be/s8mPdyEnvX8>

17. Git Commit History

- **Repository:** `namdin05/ST-Group`
- **Branch:** ``23127430-HW04``
- **Export:** ``git_commit_log.txt``
- The real history contains qualifying `.spec.ts` commits on four distinct days but only 4 qualifying commits. It does not yet meet the required minimum of 8; documentation-only commits do not count.

18. Test Summary

Metric	Value
Number of features	3
Designed test cases	46
Automated test cases	46 (FR-01: 18; FR-07: 14; FR-13: 14)
Executed test cases	46 unique cases / 138 primary browser executions
Passed executions	54 (FR-01: 21; FR-07: 9; FR-13: 24)
Failed executions	84 (FR-01: 33; FR-07: 33; FR-13: 18)

Metric	Value
Browser runs	9 primary feature-browser runs; diagnostic/invalid reruns excluded
Human-confirmed issues	17 (FR-01: 9; FR-07: 7; FR-13: 1)
Task 2 demo video	YouTube
Agent Skill video	YouTube

19. Conclusion

FR-01, FR-07, and FR-13 automation is implemented and has canonical HTML evidence for Chromium, Firefox, and WebKit. Across 138 primary executions, 54 passed and 84 failed. Seventeen selected-feature issues have human-reviewed GitHub records, while unmatched failing behaviors remain candidates. Both required video links and the public repository are documented. Requirement/test-case contradictions, fixture limitations, and the shortfall of four qualifying test-script commits still require attention before submission.